QBUS6810
Statistical Learning and Data Mining
Semester 1, 2018

Group Project: Airbnb Pricing Predictions

1. Key information

Required submissions: 1) Written report (via Turnitin submission through Canvas) and 2) Predictions for the test data (through Kaggle). You should also be prepared to submit your Python code; further instructions will be posted on Canvas.

Submission format: The recommended format is pdf (submit one file).

Deadline: Friday June 8th at 4PM.

Weight: 25% of your final grade.

Groups: Complete the assignment in groups of four to five students. Please make sure to sign into your group on Canvas, which will be used for identification and assessment purposes.

Length: The text of your written report should have a maximum of 15 pages (single spaced, 11pt). Keep in mind that making good use of your audience's time is an essential business skill. Every sentence, table and figure have to count. Extraneous and/or wrong material will generally reduce your mark.

Marking and key rules:

- A separately posted rubric indicates the marking criteria for the report.
- Carefully read the requirements for each part of the assignment.
- Please follow any further instructions announced on Canvas, particularly for submissions.
- You must use Python for this assignment. It is fine to use Excel for data manipulation, though this is neither efficient nor recommended.
- The predictions for the test data on Kaggle *must* come from your own analysis in Python. An examination of the code will be conducted for verification purposes.
- Please note that it is your responsibility to be informed of the University of Sydney and Business School rules and guidelines, and to follow them.

2. Getting the data

The data is posted on the Kaggle competition page. A link providing first access to the Kaggle competition is posted below:

https://www.kaggle.com/t/6a9dbcbc6efe43c7850bc0b5df437cb3

You will then need to create a Kaggle account based on your university e-mail address to access the competition, download the data and make submissions. The link below also leads to the competition page.

http://www.kaggle.com/c/qbus6810-s1-18

Create teams on Kaggle by merging or by inviting others. Kaggle teams should be identical to the groups you formed on Canvas, and the team number should match the group number. Each student in the group must sign up and be clearly identifiable as a member of a team.

3. Problem description

Airbnb (www.airbnb.com) is a hospitality company that runs an online marketplace for renting and leasing short-term lodging. It is interested in developing a pricing service for its users that will compute a recommended price based on the features of a listing. As a consultant working for a data analytics company, you are approached by Airbnb to develop a model for predicting nightly prices of Airbnb listings based on state-of-art techniques from statistical machine learning.

You are provided with a dataset containing detailed information on a number of existing Airbnb listings in Sydney. As part of the contract, you are asked to write a report according to the instructions given below. The client will use a test set to evaluate your work.

4. Understanding the data

There are two data files posted on Kaggle: a training set and a test set. The latter omits the response values. Kaggle randomly splits the observations in the test set into validation and test cases, but you will not know which ones are which.

When you make a submission during the competition, you get a score based on the RMSE on the validation cases. These scores are displayed on the "Public Leaderboard", and provide an ongoing ranking of teams. At the end of the competition, Kaggle will rank the teams based on the test cases only, and this will be displayed on the "Private Leaderboard". Your goal is to do as well as possible on the Private Leaderboard at the end of the competition, so please be careful not to overfit the validation cases in an attempt to improve your public ranking.

Data Description:

Each row corresponds to a separate Airbnb listing in Sydney. As a consequence of using real data, a detailed description of all the variables is not available. However, the names of the variables are self-explanatory. The first column in the data provides an identifier for each listing and is included to comply with the Kaggle format. It should not be used as a predictor in the analysis. The response variable, price, is the second column in the training dataset. It gives the dollar price per night for each listing. Variables security_deposit, cleaning_fee and extra_people are also measured in dollars and correspond to surcharges. Variables latitude and longitude specify the geographic location of each property. Several variables are Boolean, with the word true recorded as "t" and false recorded as "f".

5. Written report

The purpose of the report is to describe, explain, and justify your solution to the client. You can assume that the client is trained in business analytics. The client's time is important. The client has a limited attention span. The client is not interested in minor details. Be concise and objective. Find ways to say more with less.

Requirement:

Your report must include the validation scores for at least five different sets of predictions, including your final model. You need to make a submission on Kaggle to get each validation score. The five sets of predictions should all come from different statistical learning methods. You do not need to describe the four *additional* methods in detail.

Suggested outline:

1. Introduction: write a few paragraphs stating the business problem and summarising your final solution and results. Use plain English and avoid technical language as much as possible in this section (it should be for a wide audience).
2. Data processing and exploratory data analysis: provide key information about the data, discuss potential issues, and highlight interesting facts that are useful for the rest of your analysis.
3. Describe your process of feature engineering.
4. Methodology (your final model, your rationale for choosing it and why it makes sense for the data, description of how this model is fitted, some interpretations of the model, etc).
5. Validation set results from Kaggle and comparison of the methods.
6. Final remarks (non-technical).

6. Kaggle Competition

The purpose of the Kaggle competition is to incorporate feedback by allowing you to compare your performance with that of other groups. Participation in the competition is part of the assessment, and you must make sure that your final submission is correct. Your ranking in the competition will not directly affect your marks (apart from bonus marks, as explained below), however, we will assess whether your participation represents a genuine effort to make good predictions and improve them (in particular, you should make sure to beat the "Benchmark" score).

Real world relevance:

The ability to perform in a Kaggle competition is highly valued by employers. Some employers go as far as to set up a Kaggle competition just for recruitment.

Bonus marks:

The three teams with most accurate predictions for the test data (Private Leaderboard) will receive bonus marks for the assignment. The best performing team will receive 15 bonus marks, the second team will get 10 marks, and the third will get 5 marks. Please note that the predictions for the test data on Kaggle must come from your own analysis in Python. An examination of the code will be conducted for verification purposes.

7. Tips

- Because the number of observations in the training data is reasonably large, you may only see an improvement from using methods such as regularisation once you have a sufficient number of features (predictors) in the model.
- Several predictors have missing values. In many cases, a missing value mean that the field does not apply to that particular Airbnb listing. This is information, rather than lack of information, and you could make use of this information in your analysis.